

NORTH SYDNEY GIRLS HIGH SCHOOL

HSC Mathematics Extension 2

Assessment Task 2

Term 2 2008

Name: _____

Mathematics Class: _____

Time Allowed: 70 minutes + 2 minutes reading time

Available Marks: 53

Instructions:

- Questions are *not* of equal value.
- Start each question on a new page.
- Put your name on the top of each page.
- Attempt all three questions.
- Show all necessary working.
- Marks may be deducted for incomplete or poorly arranged work.
- Write on one side of the page only. Do not work in columns.
- Each question will be collected separately.
- If you do not attempt a question, submit a blank page with your name and the question number clearly displayed.

Question	1	2	3	Total	
E3		/20		/20	
E4	/19			/19	
E9			/14	/14	
				/53	%

Question 1: (19 marks)**Marks**

- a) i) Prove that if α is a double root of the polynomial equation $P(x) = 0$, then $P(\alpha) = P'(\alpha) = 0$. **2**
- ii) $P(x) = 12x^3 - 8x^2 - x + 1$ has a double root. Factorise $P(x)$ into irreducible factors over the real numbers. **3**
- b) i) If $1 + i$ is a zero of the polynomial $f(x) = 4x^3 - 7x^2 + 6x + 2$ explain why $1 - i$ is also a zero. **1**
- ii) Hence factorise $f(x)$ over the complex numbers. **2**
- c) The equation $x^3 + 2x - 1 = 0$ has roots α, β and γ .
- i) Find an equation with roots $\alpha^2, \beta^2, \gamma^2$. **3**
- ii) Hence evaluate $\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2$. **1**
- iii) Find an equation with roots $\alpha, -\alpha, \beta, -\beta, \gamma, -\gamma$. **2**
- d) i) Decompose $\frac{x^2 + x + 2}{(x + 2)(x^2 + 4)}$ into partial fractions over the real field. **3**
- ii) Hence show that $\int_0^2 \frac{x^2 + x + 2}{(x + 2)(x^2 + 4)} dx = \frac{3}{4} \log 2$. **2**

Question 2: (20 marks)**Start a new page****Marks**

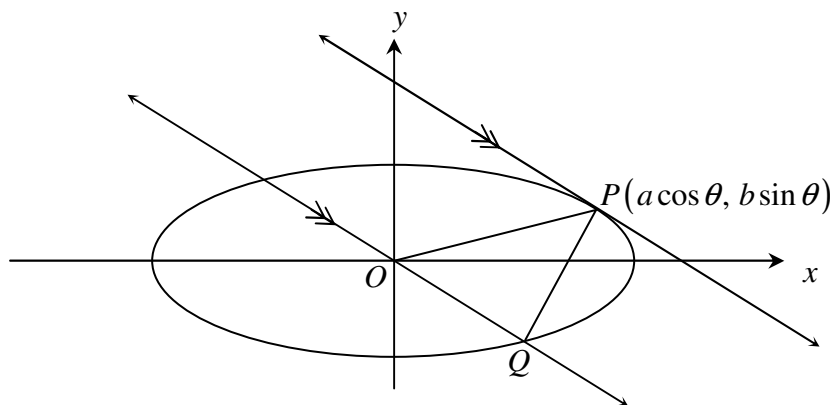
- a) Consider the hyperbola $x^2 - 4y^2 = 4$.
- i) Find
- α) the eccentricity; **1**
- β) the co-ordinates of the foci; **1**
- γ) the equations of the directrices; **1**
- ii) Hence sketch the curve indicating the asymptotes and these details. **1**

Question 2 continued:

Marks

- b) In the diagram, $P(a \cos \theta, b \sin \theta)$ is a point on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, where P is in the first quadrant.

A straight line through the origin, parallel to the tangent at P meets the ellipse at the point Q , where both P and Q lie on the same side of the y -axis.



- i) Prove that the equation of the line OQ is $bx \cos \theta + ay \sin \theta = 0$. **3**
- ii) Show that the coordinates of Q are $(a \sin \theta, -b \cos \theta)$. **3**
- iii) By considering the distance of P from OQ or otherwise, prove that the area of the triangle ΔOPQ is independent of the position of P . **3**

- b) $P\left(cp, \frac{c}{p}\right)$ and $Q\left(cq, \frac{c}{q}\right)$ where $p > 0$ and $q > 0$, are points on the rectangular hyperbola $xy = c^2$. The tangents to the rectangular hyperbola at P and Q intersect at the point R .

You are given that the tangent to the rectangular hyperbola at the point $P\left(cp, \frac{c}{p}\right)$ has the equation $x + p^2y = 2cp$. Do not prove this result.

- i) Show that the coordinates of R are $\left(\frac{2cpq}{p+q}, \frac{2c}{p+q}\right)$. **2**
- ii) P and Q are variable points on the rectangular hyperbola which move so that PQ always passes through $(2c, c)$. Show that $p + q = 2 + pq$. **3**
- iii) Find the equation of the locus of R . **2**

Question 3: (14 marks)**Start a new page****Marks**

- a) Let $P(z) = z^7 - 1$.
- i) Use de Moivre's theorem to write down the roots of $P(z) = 0$ and illustrate their relationship on an Argand diagram. **2**
 - ii) Hence write $P(z)$ as a product of real linear and quadratic factors. **2**
 - iii) Write down the quotient when $P(z)$ is divided by $z - 1$ in two different forms. **2**
 - iv) Hence show that $(1 - \cos \frac{2\pi}{7})(1 - \cos \frac{4\pi}{7})(1 - \cos \frac{6\pi}{7}) = \frac{7}{8}$. **2**
- b) In order to recognise anagrams quickly, Miss V always arranges the available letters in a circle.
- She is currently considering the letters of the word 'SURRENDER'.
- i) In how many different ways can all of the letters be arranged in a circle? **1**
 - ii) If she forms one such circular arrangement at random, what is the probability that the word 'NERDS' occurs without a break in a clockwise direction within the arrangement? **3**
 - iii) Now she selects 5 letters at random. What is the probability that the letters selected are the letters of the word 'NERDS'? **2**

End of Paper

Question 1:

- a) i) Let α be a double root of $P(x) = 0$, then $P(\alpha) = 0$.
 Also $P(x) = (x - \alpha)^2 Q(x)$ for some polynomial $Q(x)$
 Now $P'(\alpha) = (x - \alpha)^2 Q'(x) + 2(x - \alpha)Q(x)$
 $\therefore P'(\alpha) = (\alpha - \alpha)^2 Q'(\alpha) + 2(\alpha - \alpha)Q(\alpha)$
 $= 0$
 $\therefore P(\alpha) = P'(\alpha) = 0$ as required
- ii) $P(x) = 12x^3 - 8x^2 - x + 1$
 $P'(x) = 36x^2 - 16x - 1$
 $= (18x + 1)(2x - 1)$
 $\therefore$ possible double roots are $-\frac{1}{18}$ or $\frac{1}{2}$
 But 18 is not a factor of 12 so the double root must be $\frac{1}{2}$
 $\therefore P(x) = (2x - 1)^2(3x + 1)$
- b) i) $f(x) = 4x^3 - 7x^2 + 6x + 2$ has real coefficients and so complex roots will occur in conjugate pairs
 $\therefore$ as $1 + i$ is a root $1 - i$ is also a root.
- ii) Now $(x - 1 - i)(x - 1 + i) = (x - 1)^2 - i^2$
 $= x^2 - 2x + 2$
 $\therefore f(x) = 4x^3 - 7x^2 + 6x + 2$
 $= (x^2 - 2x + 2)(4x + 1)$
 $= (x - 1 - i)(x - 1 + i)(4x + 1)$
- c) Let $P(x) = x^3 + 2x - 1 = 0$ have roots $x = \alpha, \beta$ and γ .
- i) For an equation with roots $y = \alpha^2, \beta^2, \gamma^2$ the relationship is $y = x^2$
 Then $x = \sqrt{y}$
 $\therefore P(x) = 0$
 becomes $P(\sqrt{y}) = 0$
 $(\sqrt{y})^3 + 2(\sqrt{y}) - 1 = 0$
 $y\sqrt{y} + 2\sqrt{y} - 1 = 0$
 $\sqrt{y}(y + 2) = 1$
 $\sqrt{y} = \frac{1}{y + 2}$
 $y = \frac{1}{y^2 + 4y + 4}$
 $\therefore y^3 + 4y^2 + 4y - 1 = 0$
- ii) $\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2$ is the sum of the roots in pairs from i) above
 $\therefore \alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2 = \frac{c}{a}$
 $= 4$

- iii) If $ax^3 + bx^2 + cx + d = 0$ has roots α, β and γ , an equation with roots $-\alpha, -\beta, -\gamma$ is $ax^3 - bx^2 + cx - d = 0$.
 $\therefore x^3 + 2x + 1 = 0$ has roots $-\alpha, -\beta, -\gamma$
 and $(x^3 + 2x - 1)(x^3 + 2x + 1) = 0$ has roots $\alpha, \beta, \gamma, -\alpha, -\beta, -\gamma$

$$\begin{aligned}(x^3 + 2x - 1)(x^3 + 2x + 1) &= 0 \\ (x^3 + 2x)^2 - 1 &= 0 \\ x^6 + 8x^4 + 4x^2 - 1 &= 0\end{aligned}$$

b) i)
$$\frac{x^2 + x + 2}{(x+2)(x^2+4)} = \frac{A}{x+2} + \frac{Bx+C}{x^2+4}$$

$$x^2 + x + 2 = A(x^2 + 4) + (Bx + C)(x + 2)$$

If $x = 2$: $4 - 2 + 2 = 8A$

$$A = \frac{1}{2}$$

If $x = 0$: $2 = 4A + 2C$

$$2 = 2 + 2C$$

$$C = 0$$

If $x = 1$: $1 + 1 + 2 = 5A + (B + C)3$

$$4 = \frac{5}{2} + 3B$$

$$3B = \frac{3}{2}$$

$$B = \frac{1}{2}$$

$$\therefore \frac{x^2 + x + 2}{(x+2)(x^2+4)} = \frac{1}{2(x+2)} + \frac{x}{2(x^2+4)}$$

ii)
$$\begin{aligned}\int_0^2 \frac{x^2 + x + 2}{(x+2)(x^2+4)} dx &= \int_0^2 \frac{1}{2(x+2)} dx + \int_0^2 \frac{x}{2(x^2+4)} dx \\ &= \frac{1}{2} \int_0^2 \frac{1}{(x+2)} dx + \frac{1}{2} \cdot \frac{1}{2} \int_0^2 \frac{2x}{(x^2+4)} dx \\ &= \left[\frac{1}{2} \ln(x+2) + \frac{1}{4} \ln(x^2+4) \right]_0^2 \\ &= \frac{1}{2} \ln 4 + \frac{1}{4} \ln 8 - \left(\frac{1}{2} \ln 2 + \frac{1}{4} \ln 4 \right) \\ &= \frac{1}{2} \ln 4 + \frac{1}{4} \ln 8 - \frac{1}{2} \ln 2 - \frac{1}{4} \ln 4 \\ &= \frac{1}{2} \ln 2^2 + \frac{1}{4} \ln 2^3 - \frac{1}{2} \ln 2 - \frac{1}{4} \ln 2^2\end{aligned}$$

$$\begin{aligned}
 &= \ln 2 + \frac{3}{4} \ln 2 - \frac{1}{2} \ln 2 - \frac{1}{2} \ln 2 \\
 &= \frac{3}{4} \ln 2 \quad \text{as required}
 \end{aligned}$$

Question 2:

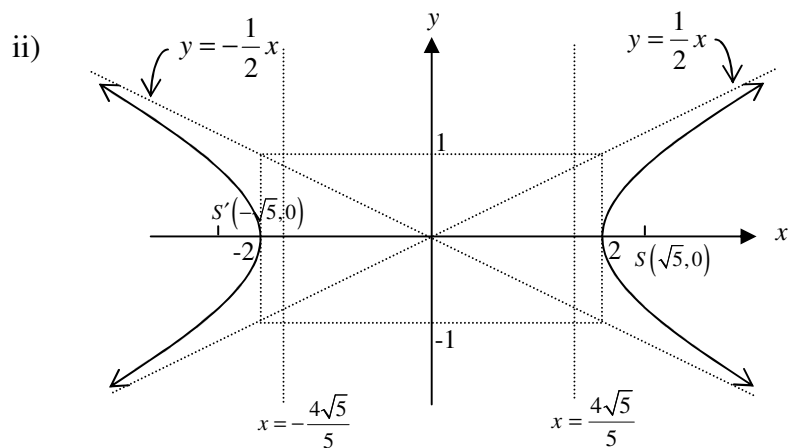
a) $x^2 - 4y^2 = 4$ becomes $\frac{x^2}{4} - y^2 = 1$.

i) $\alpha)$ $a = 2$ and $b = 1$
 $b^2 = a^2(e^2 - 1)$
 $1 = 2^2(e^2 - 1)$
 $\frac{1}{4} = e^2 - 1$
 $e^2 = \frac{5}{4}$
 $e = \frac{\sqrt{5}}{2}$ as $e > 0$

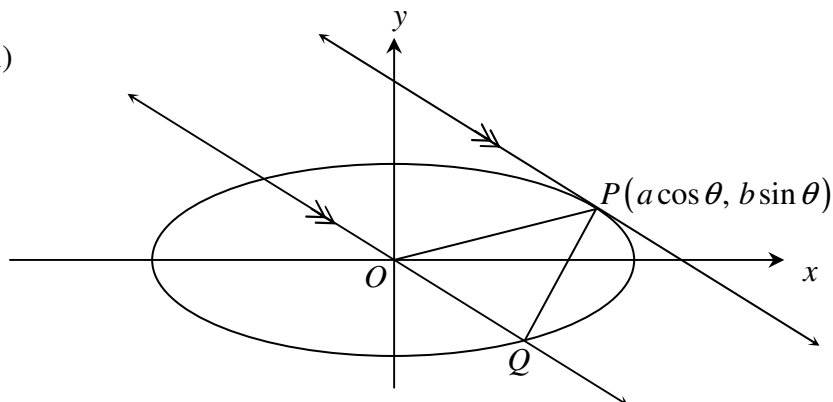
$\beta)$ $ae = \sqrt{5} \therefore$ foci are at $(\pm\sqrt{5}, 0)$

$\gamma)$
 $\frac{a}{e} = \frac{2}{\frac{\sqrt{5}}{2}}$
 $= \frac{4}{\sqrt{5}}$
 $= \frac{4\sqrt{5}}{5}$

$\therefore$ the equations of the directrices are $x = \pm \frac{4\sqrt{5}}{5}$



b) i)



At P : $x = a \cos \theta$ and $y = b \sin \theta$

$$\begin{aligned} \therefore \frac{dy}{dx} &= \frac{dy}{d\theta} \cdot \frac{d\theta}{dx} \\ &= b \cos \theta \times \frac{1}{-a \sin \theta} \\ &= -\frac{b \cos \theta}{a \sin \theta} \end{aligned}$$

$\therefore$ equation of the line OQ is

$$y = -\frac{b \cos \theta}{a \sin \theta}(x)$$

$$ay \sin \theta = -bx \cos \theta$$

$$\therefore bx \cos \theta + ay \sin \theta = 0 \quad \text{as required}$$

ii) For Q substitute $y = -\frac{b \cos \theta}{a \sin \theta}(x)$ into $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$$\text{Then } \frac{x^2}{a^2} + \frac{b^2 x^2 \cos^2 \theta}{b^2 a^2 \sin^2 \theta} = 1$$

$$x^2 \left(1 + \frac{\cos^2 \theta}{\sin^2 \theta} \right) = a^2$$

$$x^2 \left(\frac{\sin^2 \theta + \cos^2 \theta}{\sin^2 \theta} \right) = a^2$$

$$x^2 = a^2 \sin^2 \theta$$

$$x = \pm a \sin \theta \quad \text{but } x > 0$$

$$x = a \sin \theta$$

Substituting into OQ gives $y = -b \cos \theta$

$\therefore$ the coordinates of Q are $(a \sin \theta, -b \cos \theta)$

iii) The distance of $P(a \cos \theta, b \sin \theta)$ from OQ : $bx \cos \theta + ay \sin \theta = 0$ is

$$\begin{aligned} h &= \frac{|b(a \cos \theta) \cos \theta + a(b \sin \theta) \sin \theta|}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}} \\ &= \frac{ab(\cos^2 \theta + \sin^2 \theta)}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}} \quad \text{as } a, b > 0 \\ &= \frac{ab}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}} \end{aligned}$$

The distance OQ is $b = \sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}$

$$\therefore A = \frac{1}{2}bh$$

$$A = \frac{1}{2} \cdot \frac{ab}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}} \cdot \sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}$$

$$= \frac{ab}{2} \quad \text{which is independent of } \theta \text{ and consequently the position of } P$$

c) i) At P : $x + p^2 y = 2cp$ (1)
 At Q : $x + q^2 y = 2cq$ (2)
 (1) - (2): $(p^2 - q^2)y = 2c(p - q)$ but $p \neq q$

$$\therefore y = \frac{2c}{p + q}$$

Substitute into (1): $x + p^2 \left(\frac{2c}{p + q} \right) = 2cp$

$$x = \frac{2cp(p + q) - 2cp^2}{p + q}$$

$$= \frac{2cpq}{p + q}$$

$$\therefore R = \left(\frac{2cpq}{p + q}, \frac{2c}{p + q} \right) \quad \text{as required}$$

ii) $m_{PQ} = \frac{\frac{c}{p} - \frac{c}{q}}{cp - cq}$

$$= \frac{q - p}{pq(p - q)}$$

$$= -\frac{1}{pq}$$

$\therefore$ equation PQ : $y - \frac{c}{p} = -\frac{1}{pq}(x - cp)$

$$pqy - cq = -x + cp$$

$$x + pqy = c(p + q)$$

This passes through $(2c, c)$

$$\therefore 2c + pqc = c(p + q)$$

ie: $2 + pq = p + q$ as required

iii) At R : $x = \frac{2cpq}{p + q}$ (1)

$$y = \frac{2c}{p + q}$$
 (2)

Also $2 + pq = p + q$ (3)

From (1) and (2): $x = pqy$

$$\therefore pq = \frac{x}{y}$$

From (2): $p + q = \frac{2c}{y}$

Substituting both these into (3): $2 + \frac{x}{y} = \frac{2c}{y}$
 $2y + x = 2c$

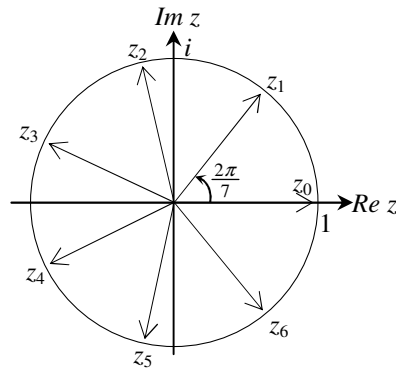
$\therefore$ the locus of R is: $x + 2y - 2c = 0$

Question 3:

a) i) $P(z) = z^7 - 1 = 0$

$$z^7 = 1$$

$$z = \cos\left(\frac{2n\pi}{7}\right) + i \sin\left(\frac{2n\pi}{7}\right) \text{ for } n = 0, 1, 2, 3, 4, 5, 6$$



ii) By symmetry, $z_1 = \bar{z}_6$; $z_2 = \bar{z}_5$; $z_3 = \bar{z}_4$

$$\therefore (z - z_1)(z - z_6) = z^2 - 2z \operatorname{Re} z_1 + 1$$

$$(z - z_2)(z - z_5) = z^2 - 2z \operatorname{Re} z_2 + 1$$

$$(z - z_3)(z - z_4) = z^2 - 2z \operatorname{Re} z_3 + 1$$

Also, $z_0 = 1$

$$\begin{aligned} \therefore z^7 - 1 &= (z - z_0)(z - z_1)(z - z_2)(z - z_3)(z - z_4)(z - z_5)(z - z_6) \\ &= (z - 1)(z^2 - 2z \operatorname{Re} z_1 + 1)(z^2 - 2z \operatorname{Re} z_2 + 1)(z^2 - 2z \operatorname{Re} z_3 + 1) \\ &= (z - 1)(z^2 - 2z \cos \frac{2\pi}{7} + 1)(z^2 - 2z \cos \frac{4\pi}{7} + 1)(z^2 - 2z \cos \frac{6\pi}{7} + 1) \end{aligned}$$

iii) $\frac{z^7 - 1}{z - 1} = z^6 + z^5 + z^4 + z^3 + z^2 + z + 1$ and

$$\frac{z^7 - 1}{z - 1} = (z^2 - 2z \cos \frac{2\pi}{7} + 1)(z^2 - 2z \cos \frac{4\pi}{7} + 1)(z^2 - 2z \cos \frac{6\pi}{7} + 1)$$

iii) Now

$$z^6 + z^5 + z^4 + z^3 + z^2 + z + 1 = (z^2 - 2z \cos \frac{2\pi}{7} + 1)(z^2 - 2z \cos \frac{4\pi}{7} + 1)(z^2 - 2z \cos \frac{6\pi}{7} + 1)$$

Substituting $z = 1$: $7 = (1^2 - 2 \cos \frac{2\pi}{7} + 1)(1^2 - 2 \cos \frac{4\pi}{7} + 1)(1^2 - 2 \cos \frac{6\pi}{7} + 1)$

$$\begin{aligned}
7 &= \left(1^2 - 2\cos\frac{2\pi}{7} + 1\right)\left(1^2 - 2\cos\frac{4\pi}{7} + 1\right)\left(1^2 - 2\cos\frac{6\pi}{7} + 1\right) \\
&= \left(2 - 2\cos\frac{2\pi}{7}\right)\left(2 - 2\cos\frac{4\pi}{7}\right)\left(2 - 2\cos\frac{6\pi}{7}\right) \\
&= 8\left(1 - \cos\frac{2\pi}{7}\right)\left(1 - \cos\frac{4\pi}{7}\right)\left(1 - \cos\frac{6\pi}{7}\right) \\
\text{i.e. } &\left(1 - \cos\frac{2\pi}{7}\right)\left(1 - \cos\frac{4\pi}{7}\right)\left(1 - \cos\frac{6\pi}{7}\right) = \frac{7}{8} \quad \text{as required}
\end{aligned}$$

b) S U R E N D
 R E
 R

- i) Place one letter down as a marker. Then there are 8 remaining letters, 3 of one kind and 2 of another.

$$\text{Number of arrangements} = \frac{8!}{3!2!} = 3360$$

- ii) Tie 'NERDS' together as one group and place this group as the marker. Then there are 4 remaining letters including a pair of R's

$$\text{Number of arrangements} = \frac{4!}{2!} = 12$$

$$\therefore P(\text{seeing NERDS}) = \frac{12}{3360} = \frac{1}{280}$$

- iii) $P(\text{NERDS in order}) = \frac{1}{9} \times \frac{2}{8} \times \frac{3}{7} \times \frac{1}{6} \times \frac{1}{5}$

But there are 5! Ways of ordering the letters of NERDS

$$\begin{aligned}
\therefore P(\text{NERDS in any order}) &= \frac{1}{9} \times \frac{2}{8} \times \frac{3}{7} \times \frac{1}{6} \times \frac{1}{5} \times 5! \\
&= \frac{2 \times 3 \times 5!}{9 \times 8 \times 7 \times 6 \times 5} \\
&= \frac{1}{21}
\end{aligned}$$

End of Solutions